



AI-POWERED THREAT DETECTION

Safe-Link

Citizen Fraud Shield

AI-Powered URL Security Intelligence to Protect India's Digital Citizens

ET AI Hackathon 2.0 | Problem Statement 6: Digital Public Safety | Phase 2 Submission

THE PROBLEM

India's Cybercrime Crisis



1.14M

Cybercrime complaints
in 2023 alone



₹1,776 Cr

Lost to Digital Arrest
scams in 9 months



60%

Year-over-year
increase in cybercrimes



Phishing URLs are the #1 entry point for digital fraud — yet citizens have no real-time tool to verify suspicious links before clicking.

OUR SOLUTION

Safe-Link: Citizen Fraud Shield

An AI-powered platform that analyzes any URL in real time to detect phishing, fraud, and malicious links — protecting Indian citizens before they click.



Instant URL Scan

Paste any suspicious link and get a threat verdict in seconds



AI Risk Scoring

XGBoost ML model assigns composite threat score with confidence %



5-Level Analysis

Keywords → URL expansion → Safe Browsing → Domain → ML model

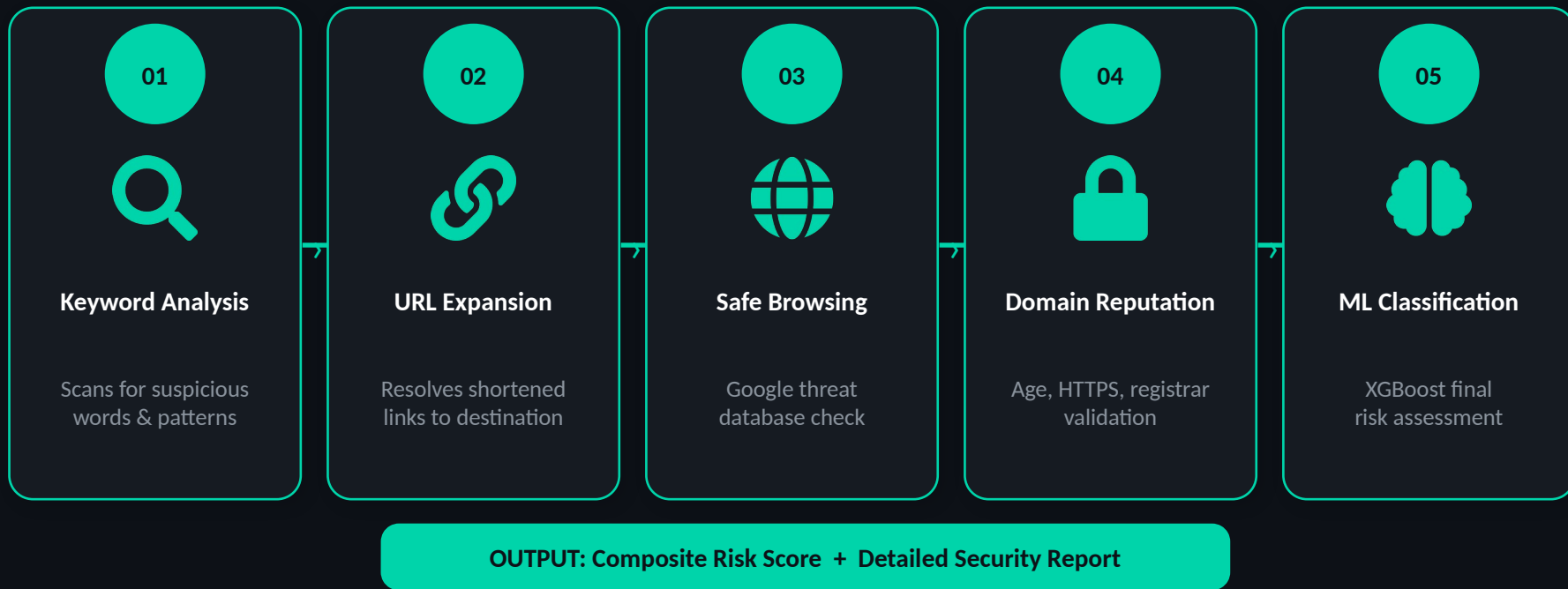


Shareable Reports

Generate & share detailed security reports for awareness

HOW IT WORKS

5-Level Deep Scan Pipeline



ARCHITECTURE

System Architecture

FRONTEND

- ▶ React + TypeScript
- ▶ Tailwind CSS
- ▶ Vite Build Tool
- ▶ Real-time UI Updates

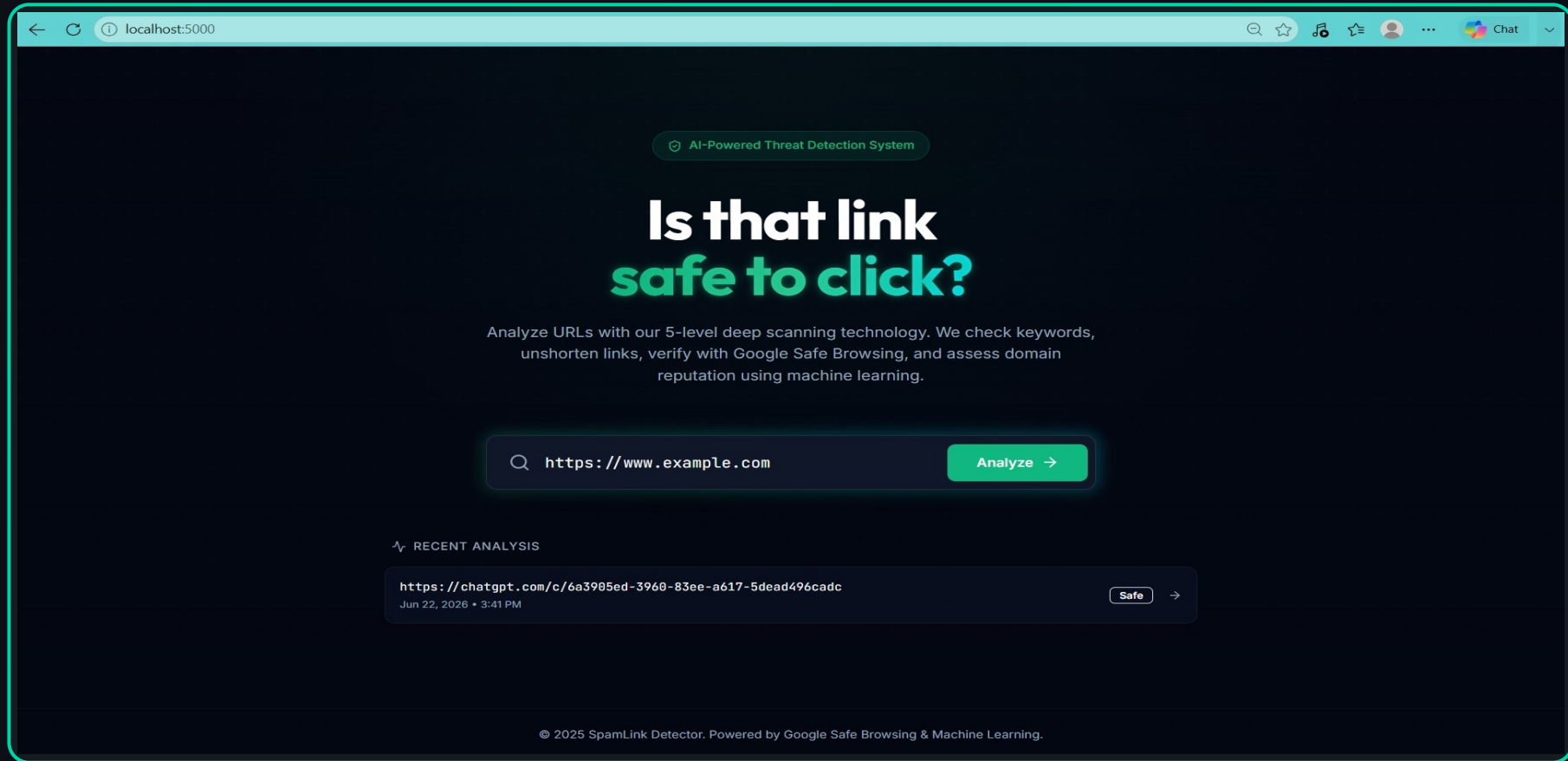
BACKEND API

- ▶ Node.js + Express
- ▶ URL Analysis Engine
- ▶ Drizzle ORM + DB
- ▶ REST API Endpoints

AI / SECURITY

- ▶ XGBoost ML Model
- ▶ Google Safe Browsing
- ▶ Domain Intelligence
- ▶ Threat Scoring Engine





← ↻ ⓘ localhost:5000

🔍 ☆ 🎵 📖 👤 ... Chat ▾

🛡️ AI-Powered Threat Detection System

Is that link safe to click?

Analyze URLs with our 5-level deep scanning technology. We check keywords, unshorten links, verify with Google Safe Browsing, and assess domain reputation using machine learning.

🔍

Analyze →

📄 RECENT ANALYSIS

🔗 <https://www.example.com>

Jun 22, 2026 • 4:12 PM

Safe →

🔗 <https://chatgpt.com/c/6a3985ed-3960-83ee-a617-5dead496cad0>

Jun 22, 2026 • 3:41 PM

Safe →

© 2025 SpamLink Detector. Powered by Google Safe Browsing & Machine Learning.

LIVE DEMO — Scan Result: SAFE URL (Risk Score: 0)

← ↻ ⓘ localhost:5000/scans/2 🔍 ☆ 🎵 ⚙️ 👤 ⋮ 🗨️ Chat ▾

← Back to Scanner

SCAN ID: 000002

SAFE June 22nd, 2026 4:12 PM

https://www.example.com

Visit URL ↗ Share Report

0 RISK SCORE

Composite Threat Score

Keyword Analysis ✓

Scans for suspicious words and patterns

Suspicious Keywords: None

URL Length: 23 chars

Symbols: Normal

URL Expansion ⓘ

Resolves shortened links to destination

Shortened: No

Google Safe Browsing ✓

Checks against global threat database


No Threats Found
URL is not in Google's blacklist

Domain Reputation ✓

Evaluates age, HTTPS, and registrar

HTTPS: Valid Certificate

Domain Age: Unknown

Registrar: Hidden

ML Classification ✓

Final XGBoost risk assessment

Confidence Score 0.0%

Based on keyword patterns, URL structure, domain reputation, and external threat intelligence feeds.

LIVE DEMO — Scan Result: SUSPICIOUS URL (Risk Score: 45)

localhost:5000/scans/3

Chat

[← Back to Scanner](#)SCAN ID: 000003

LOW RISK

June 22nd, 2026 4:13 PM

http://www.login-support.example/help

Visit URL

Share Report

45

RISK SCORE

Composite Threat Score

Keyword Analysis

Scans for suspicious words and patterns

Suspicious Keywords: **login**

URL Length: 37 chars

Symbols: Normal

⚠️

URL Expansion

Resolves shortened links to destination

Shortened: No

ℹ️

Google Safe Browsing

Checks against global threat database

✓

No Threats Found

URL is not in Google's blacklist

✓

Domain Reputation

Evaluates age, HTTPS, and registrar

HTTPS: **Missing / Invalid**

Domain Age: Unknown

Registrar: Hidden

⚠️

ML Classification

Final XGBoost risk assessment

Confidence Score 45.0%

Based on keyword patterns, URL structure, domain reputation, and external threat intelligence feeds.

✓

TECH STACK

Technologies Used

Frontend

React • TypeScript • Tailwind CSS • Vite

Backend

Node.js • Express.js • TypeScript • Drizzle ORM

AI / ML

XGBoost Classifier • Feature Engineering • Risk Scoring Engine

Security APIs

Google Safe Browsing API • Domain Intelligence • URL Resolver

DevOps

GitHub • Replit • REST API Architecture

IMPACT

Business Impact & Scale

900M+

Internet users in India — our potential protection reach

24/7

Real-time protection with zero manual effort

5-Layer

Defense — more thorough than any single API check

Who benefits:

► **General Citizens** — Verify suspicious links from WhatsApp, SMS, email

► **Senior Citizens** — Protection against digital arrest scam links

► **Bank Customers** — Detect fake banking & KYC phishing URLs

► **Enterprises** — Employee security training and threat awareness

FUTURE SCOPE

Roadmap: Expanding the Shield

V2.0

Digital Arrest Scam Detector

Detect fake CBI/ED/Customs portal URLs and govt impersonation domains in real time

V2.1

WhatsApp Link Scanner Bot

Telegram/WhatsApp bot integration so citizens can forward suspicious links directly for instant scan

V3.0

Fraud Network Graph Intelligence

Graph AI to map connected phishing domains, track scam campaigns, and identify fraud networks

V3.1

12-Language Citizen Advisory

AI explanations in Telugu, Hindi, Tamil, Kannada & more — making safety accessible to all Indians



Safe-Link

Protecting India's Digital Citizens — One URL at a Time

GitHub: github.com/Divya18n/Safe-Link | Problem Statement 6: Digital Public Safety

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